

Genetics Basics & Mendel's Experiments

Worksheet · PU-II Biology, Chapter 4

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Worksheet – Genetics Basics & Mendel's Experiments

15 questions · answer key on the last page

Q1. Who is known as the "father of genetics"?

- (a) Charles Darwin
- (b) Gregor Johann Mendel
- (c) Thomas Hunt Morgan
- (d) Hugo de Vries

Q2. Mendel conducted his hybridisation experiments on which plant?

- (a) Maize
- (b) Garden pea (*Pisum sativum*)
- (c) Sunflower
- (d) Wheat

Q3. What did Mendel call the functional unit of inheritance that carries information to express a trait?

- (a) Chromosome
- (b) Allele
- (c) Factor
- (d) Locus

Q4. The genetic constitution of an organism is called its:

- (a) Phenotype
- (b) Genotype
- (c) Homozygous state
- (d) Heterozygous state

Q5. The external, observable appearance of an organism (e.g. tall or dwarf) is called its:

- (a) Genotype
- (b) Phenotype
- (c) Locus
- (d) Allele

Q6. An organism with two identical alleles for a gene (e.g. TT or tt) is said to be:

- (a) Heterozygous
- (b) Homozygous
- (c) Hybrid
- (d) Hemizygous

Q7. An organism with two dissimilar alleles for a gene (e.g. Tt) is said to be:

- (a) Homozygous
- (b) Heterozygous
- (c) Pure-breeding
- (d) True-breeding

Q8. For how many years did Mendel conduct his hybridisation experiments on garden pea?

- (a) Three
- (b) Five
- (c) Seven
- (d) Ten

Q9. Which of the following was NOT one of the seven contrasting characters studied by Mendel in garden pea?

- (a) Stem height
- (b) Seed shape

- (c) Flower colour
- (d) Leaf shape

Q10. In Mendel's pea experiments, which trait was dominant for pod colour?

- (a) Yellow
- (b) Green
- (c) Constricted
- (d) Inflated

Q11. The specific region on a chromosome where a gene resides is called its:

- (a) Allele
- (b) Locus
- (c) Genome
- (d) Genotype

Q12. A cross between two homozygous individuals differing in a single character is called a:

- (a) Dihybrid cross
- (b) Monohybrid cross
- (c) Test cross
- (d) Back cross

Q13. A cross between an F1 individual and either of its parents is termed a:

- (a) Test cross
- (b) Out cross
- (c) Back cross
- (d) Dihybrid cross

Q14. A backcross between an F1/F2 hybrid of unknown genotype and its homozygous recessive parent is called a:

- (a) Reciprocal cross
- (b) Out cross
- (c) Test cross
- (d) Trihybrid cross

Q15. A backcross between an F1/F2 hybrid of unknown genotype and a homozygous dominant parent is called:

- (a) Test cross
- (b) Monohybrid cross
- (c) Out cross
- (d) Dihybrid cross

Answer Key & Explanations

- Q1.** (b) Gregor Johann Mendel — Gregor Johann Mendel is called the father of genetics for his pioneering hybridisation experiments on garden pea using statistical analysis.
- Q2.** (b) Garden pea (*Pisum sativum*) — Mendel chose the garden pea, *Pisum sativum*, because it was easy to cultivate and had several contrasting characters.
- Q3.** (c) Factor — Mendel used the term "factor" for what we now call a gene, the functional unit of inheritance.
- Q4.** (b) Genotype — Genotype refers to the genetic makeup (e.g. TT, Tt, tt), whereas phenotype refers to the observable appearance.
- Q5.** (b) Phenotype — Phenotype is the outward physical expression of a trait, such as tall or dwarf stem height in pea.
- Q6.** (b) Homozygous — Homozygous individuals carry two similar alleles of a gene, such as TT or tt.
- Q7.** (b) Heterozygous — Heterozygous individuals carry two dissimilar alleles for a gene, such as Tt.
- Q8.** (c) Seven — Mendel's pea experiments spanned seven years, from 1856 to 1863.
- Q9.** (d) Leaf shape — Mendel's seven characters were stem height, seed shape, seed colour, pod shape, pod colour, flower colour, and flower position; leaf shape was not among them.
- Q10.** (b) Green — For pod colour, green was dominant over yellow, as recorded in Mendel's seven contrasting character pairs.
- Q11.** (b) Locus — A locus (plural loci) is the specific location on a chromosome occupied by a gene or allele.
- Q12.** (b) Monohybrid cross — A monohybrid cross involves parents differing in only one contrasting character, unlike a dihybrid cross which involves two.
- Q13.** (c) Back cross — A back cross is simply a cross between an F1 individual and one of its two parents, regardless of that parent's genotype.
- Q14.** (c) Test cross — A test cross specifically uses a homozygous recessive parent to reveal whether the hybrid is homozygous or heterozygous dominant.
- Q15.** (c) Out cross — An out cross uses a homozygous dominant parent, unlike a test cross which uses a homozygous recessive parent.